

WhisperWard OSINT: Case Study

An explainable digital investigations platform for Trust & Safety and child-protection investigations.

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Live demo: whisperward-osint.onrender.com **License:** AGPL-3.0

Overview

WhisperWard is a deployed, open-source intelligence (OSINT) platform designed to support investigations into online child exploitation on gaming and social platforms. I designed and built the entire system end-to-end. It guides investigators from initial subject discovery through to court-defensible evidence packaging by combining automated public data collection, cross-platform identity correlation, analyst-gated identity resolution, a fully transparent weighted risk-scoring engine, and forensic evidence handling, all backed by mandatory human review at its core.

This is not a prototype. WhisperWard runs a live public demonstration environment, carries 594 automated tests in continuous integration, and is published under AGPL-3.0 so its safeguards remain open and auditable. Since first release it has been evolved through a four-phase platform refactor: the investigation machinery now lives in a reusable core, with child safety as its first specialization module, and the platform speaks industry standards through STIX 2.1 export and an honest MITRE ATT&CK mapping. Most recently it gained a persistence layer that tracks account reconstitution — the pattern account-level enforcement does not capture.



WhisperWard boot screen — operator shown as unauthenticated until sign-in, with the live active-case count

The Problem

Online child exploitation on platforms like Roblox and Discord continues to grow. These environments are primary spaces for children to socialize, and predators heavily target them for grooming.

Investigators in law enforcement, platform Trust & Safety, and nonprofits face significant challenges: grooming signals are fragmented across platforms and time, and existing tools often lack the ability to correlate data and produce defensible, explainable assessments.

Four core constraints shaped WhisperWard's architecture:

Fragmented, cumulative signals. Grooming is rarely evident in a single message. It emerges as patterns such as age probing, boundary testing, secrecy requests, and platform migration, unfolding across time and accounts. The tooling must reason about the pattern, not the message.

Explainable, defensible assessments. Any system that may inform referrals or legal processes must let reviewers trace every risk point to its source. A black-box score is unacceptable in this domain.

Human authority over identity claims. Saying two accounts are the same person is a consequential claim. The machine may propose it with evidence; only a named analyst may resolve it, and that decision must be permanently attributable.

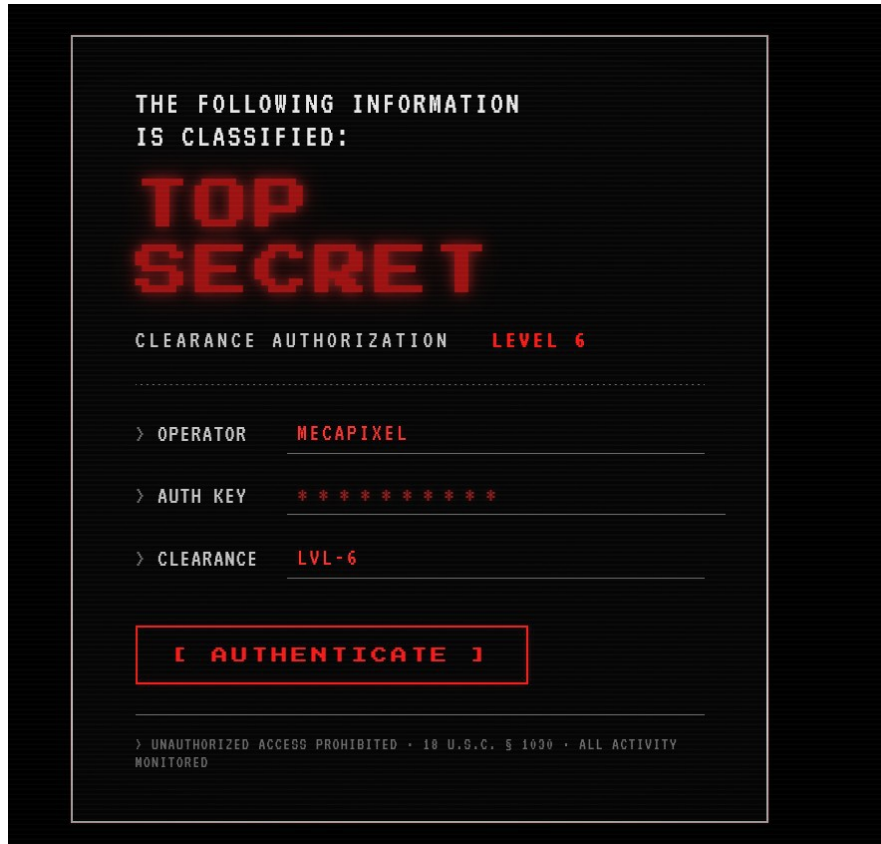
Enforcement measured at the operator, not the account. Removing an account ends that account, not the behavior behind it. An operator who returns within days has defeated the enforcement while the record shows a successful takedown. Measuring disruption requires modeling which account replaced which, and how quickly.

What WhisperWard Does

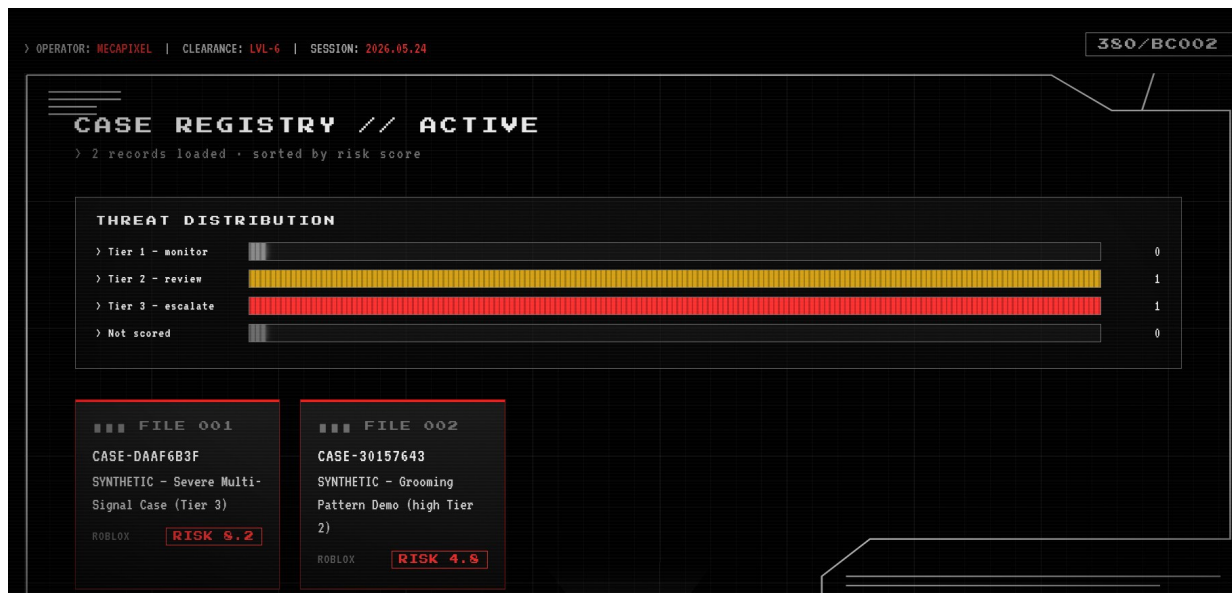
The platform executes a complete investigative pipeline:

- 1. Collection.** Gathers public profile data from platforms like Roblox and correlates usernames across sites.
- 2. Correlation.** Links identities using fuzzy matching, writing-style analysis, and perceptual avatar hashing, exposed through a first-class CLI correlate command.
- 3. Identity resolution.** Proposes candidate entities from correlation output with full per-membership justification; promotion to a resolved identity requires a named analyst and is sealed into the chain of custody. A justified identity graph and a strictly reconstructive investigation timeline sit alongside it.
- 4. Persistence tracking.** Orders removed and successor accounts into directed reconstitution chains, with the return interval on every link, so enforcement outcomes can be measured rather than assumed.
- 5. Behavioral analysis.** Detects grooming patterns in chat content, accounting for multi-step sequences.
- 6. Risk scoring.** Generates a weighted, fully explainable risk score with tier recommendations, reasoning over graph-corroborated platforms when identity-graph inputs are present.
- 7. Evidence & reporting.** Packages artifacts with SHA-256 hashing and chain-of-custody manifests, supports redaction and referral export, and exports cases as byte-stable STIX 2.1 bundles with an ATT&CK technique mapping.

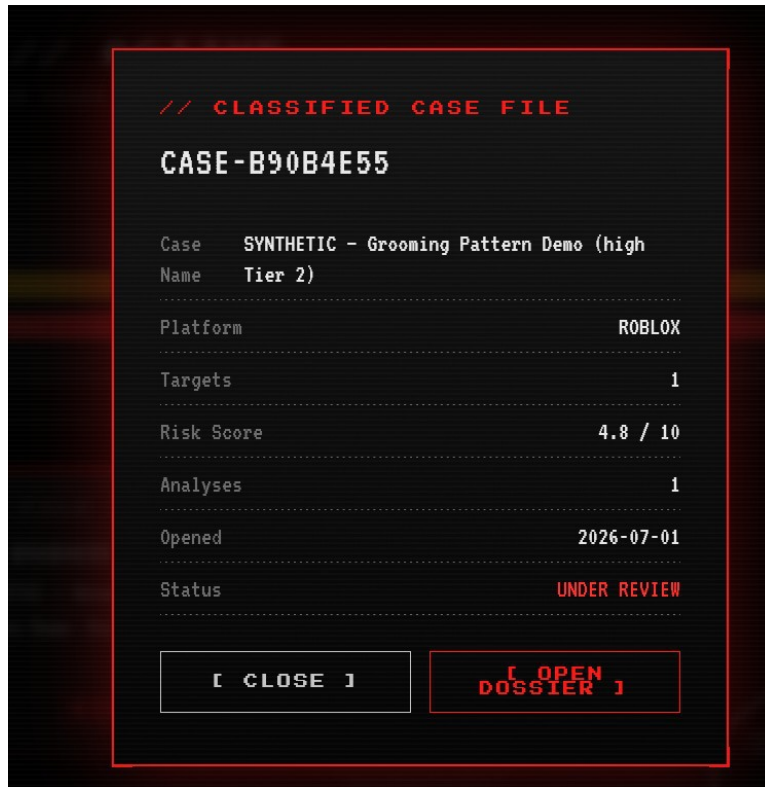
8. Human review. Every escalation requires human approval. The system provides evidence and reasoning. It never makes autonomous decisions.



Operator authentication — a signed session is required before any case data is shown



Case registry — active cases sorted by risk, with the D3 threat-tier distribution



Classified case file — quick-view popup

The Explainable Risk Engine (Core Differentiator)

The risk engine is the heart of WhisperWard. Instead of opaque scores, it evaluates subjects across five weighted components, each with traceable contributions and plain-language explanations:

Component	Weight	What it measures
Grooming language	40%	Multi-category pattern detection in chat content
Cross-platform correlation	25%	Same identity detected across multiple platforms
Anonymization	15%	Use of Tor / VPN infrastructure
Behavioral velocity	10%	Account age and rapid friend acquisition
Historical signals	10%	Prior flags in the investigation database

A synergy bonus rewards corroborating signals. Scores map to three tiers: Tier 1 (Monitor), Tier 2 (Human Review Required), and Tier 3 (Escalate). The Tier 3 threshold is deliberately set above what any single-platform signal can reach on its own, so escalation requires genuine cross-platform or historical corroboration.

When identity-graph inputs are present, the cross-platform component reasons over graph-corroborated platforms, and a contradiction on any supporting edge caps confidence without ever altering the score itself — uncertainty is surfaced, never hidden inside the number.

The result: every score is fully accountable. A reviewer sees not “8.2,” but “8.2 because grooming language contributed 4.0 points, cross-platform presence 2.5, behavioral velocity 0.7,” each with its own plain-language explanation.



Risk breakdown panel showing weighted components and the terminal intel log

From Tool to Investigation Platform

After first release, WhisperWard was deliberately re-architected across four platform phases, growing the test suite from 424 to 594 along the way, green at every milestone.

Core extraction. The investigation machinery — contracts, registry, plugin base, correlation, scoring, evidence, redaction, retention, and reporting — was separated into a reusable core. Child safety became the first specialization module, registering its collectors, classifiers, and behavioral indicators through a registration seam. The core never imports a specialization, so the same engine can support other Trust & Safety domains.

Evidence and explainability engine. Every score component and result carries a confidence level with enumerated reasons that never alter the score. Findings are promoted to queryable explanation objects linked to their evidence, analysts annotate through an append-only notes layer chained into the custody log, and the full case lifecycle leaves a tamper-evident audit trail.

Entity resolution and identity graph. The resolver proposes candidate identities from correlation output; a contradiction on any supporting edge excludes an account with the reason recorded. Promotion requires a named analyst and lands in the chain of custody — unattributed resolution refuses to run. The identity graph carries a complete justification on every edge, answers path queries with the evidence behind them, and serializes deterministically for evidence packaging. The

investigation timeline is strictly reconstructive: every event names its source, and no event is ever inferred.

Standards-aware intelligence. Cases export as byte-stable STIX 2.1 bundles — user-account observables, correlation relationships with confidence and rationale, analyst-resolved entities, and a suspicious-activity grouping — that deliberately never emit adversary-assertion object types. The MITRE ATT&CK mapping is honest by construction: it maps only fired technical signals, grades each mapping direct or analogue, and states which findings fall outside the framework's scope and where they are documented instead. A written threat model covers the platform's own assets, adversaries, failure modes, and residual risks.

Measuring Enforcement: The Persistence Graph

Correlation answers whether two accounts appear to be operated by the same person. It does not answer whether one replaced the other after a removal — and that directed, temporal relationship is what shows an enforcement action failed. The identity graph could not express it: it stores edges under a sorted pair, so direction is discarded by design.

The persistence layer sits on top of that graph rather than replacing it. It holds a directed graph of succession and cross-platform handoff edges, reusing the identity graph's justification model so an edge without a rationale cannot exist at all.

Direction is derived, not asserted. Order comes from removal and creation timestamps, never from the correlation, which carries no ordering. Cycles are refused, since a succession chain is temporally ordered and a cycle means the ordering is wrong.

Inference is held to a floor. Succession is inferred from correlation output only above a configurable strength threshold and inside a reconstitution window, skipping any pair carrying a contradiction. Every derived link is flagged as inferred, so analyst-observed successions stay separable from suggested ones.

Return intervals are the measurement. Each link records how long the operator took to come back, with a median per chain. A return measured in hours is not a coincidence, and it is the number that makes a pattern legible to a platform.

Two export views. A full-fidelity referral view retains identifiers for the lawful recipient. A platform-facing view replaces every account with an opaque chain and position label while preserving chain length, intervals, and platform span — carrying an explicit note that chains are correlation groupings, not identifications of natural persons.

This is the first component of the platform that is not an implementation of an established technique. Reconstitution chains with return intervals are an original analytic contribution, built to answer a question the existing tooling does not ask.

A Defect the Test Suite Could Not See

Wiring the persistence layer to the command line produced a result worth recording. With the module complete and its full test suite green, the first run against real case data inferred nothing — at any threshold, with no error explaining why.

The cause was a sentinel mismatch. The correlation engine records a clean comparison by writing the sentence “no contradictions detected” into its contradiction field, while every downstream consumer tests that field for emptiness. A clean pair therefore read as contradicted, and every pair was discarded. The unit tests passed throughout, because they construct their own justifications using the empty-string convention rather than the engine's real output — the defect lived precisely in the gap between the two.

Tracing it through version history showed the sentinel was introduced with the extracted correlation engine and became live one day later, when the identity graph first consumed the field — roughly eight weeks before discovery. The effect was confined to confidence labeling: cross-platform corroboration was capped at medium and could never reach high, with an inaccurate reason recorded on the case.

The failure direction matters. The platform understated its own evidentiary support rather than overstating it, which is the correct direction to fail in work that may inform a referral. The fix was scoped to the consuming module with tests covering the sentinel directly; correcting the engine itself was deliberately deferred to an isolated change, because it moves confidence labels across every existing case and that movement should be visible on its own rather than bundled into a feature commit.

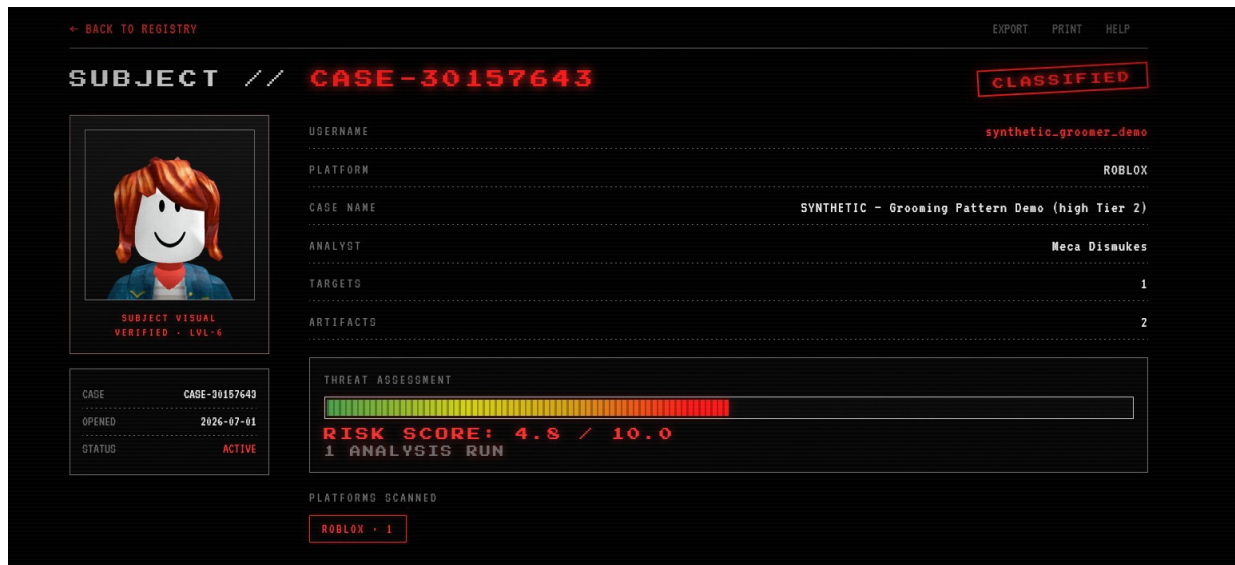
The broader lesson is recorded here on purpose: a green suite demonstrates that components behave as their tests assume, not that they agree with each other. Integration against real data remains the only check on the assumptions the tests themselves encode.

Demonstration

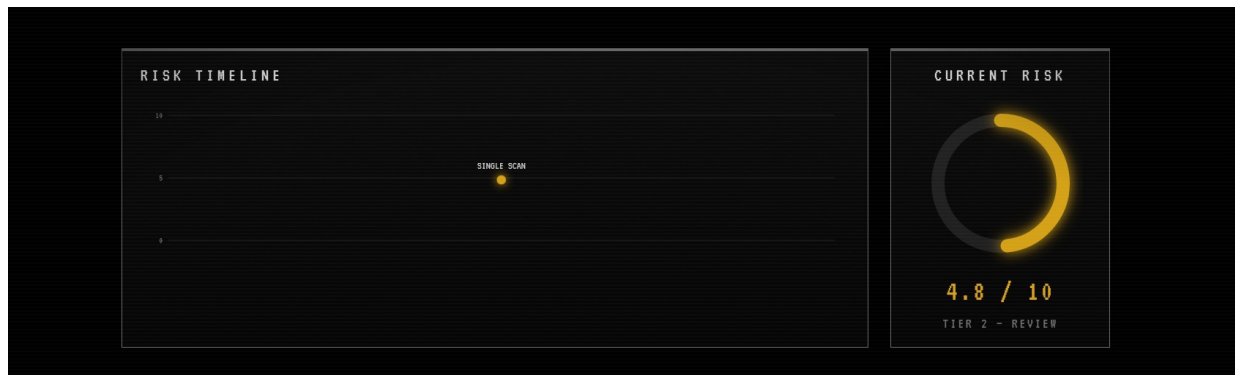
The live demo carries six active cases. All use investigator-created test accounts and fabricated data, which the project's governance framework permits — no real minor's data and no third-party accounts are processed. Four are burner accounts created for testing; the two synthetic demo cases below were built from that same fabricated material to exercise the engine across its full range and show that it discriminates severity based on real signals rather than flagging everything alike.

Tier 2, Human Review Required (score 4.8). A synthetic subject exhibiting a grooming-language pattern (age probing, a compliment, platform-migration pressure, and a secrecy request) across two platforms on a young account. The engine flags it for review without over-escalating, which is the realistic, common case.

Tier 3, Escalate (score 8.2). A synthetic subject exhibiting a strong multi-category grooming pattern across four platforms on a very new, high-velocity account. Every axis fires, the corroboration synergy applies, and the engine escalates to an evidence package with mandatory human sign-off.



Synthetic case dossier (Tier 3, score 8.2) with explainable scoring



Risk timeline and current-risk gauge

The contrast between the two demonstrates the design goal: the score is not a gut feeling, it is a defensible sum of documented parts, and the same engine correctly separates a review-level case from an escalation-level one.

Governance, Ethics, and Safety

Safeguards are foundational, not afterthoughts:

Human-in-the-loop by design. No autonomous decisions. Every escalation requires human sign-off before any action, and resolving two accounts into one identity requires a named analyst whose decision is sealed into the custody chain.

Approval-gated CSAM handling. Perceptual-hash detection is disabled by default and gated behind explicit approval and credentials available only to vetted organizations. It never stores image content, processes hashes in memory only, and flags every match for mandatory human review.

Synthetic data only. All demonstration and testing uses clearly labeled fabricated data and investigator-created test accounts. No real minor's data and no third-party accounts are processed.

Enforced operator identity. The web dashboard requires an authenticated operator session, signed and verified server-side, before any case page is displayed, and the authoritative operator record lives in the tamper-evident chain-of-custody log.

No adversary assertions in exports. The STIX 2.1 export shares observables, relationships, and analyst-resolved entities with their confidence and rationale — it never emits object types that would assert someone is a threat actor. That determination belongs to humans and due process.

Persistence chains are groupings, not identifications. Reconstitution output carries an explicit statement that a chain describes accounts believed to be operated in coordination, and that inferred links are suggestions for an analyst to confirm. The platform-facing view withholds identifiers entirely.

Open and auditable. Published under AGPL-3.0 specifically so the safeguards cannot be quietly stripped out.

Engineering

Stack. Python, FastAPI, SQLite, D3.js, NetworkX, and stix2, with a modular, plugin-based collection architecture.

Quality. 594 automated tests running in GitHub Actions on every push. The scoring engine, behavioral classifier, correlation, entity resolution, identity graph, persistence graph, and STIX export modules are independently tested, and the suite runs clean from a fresh clone.

Architecture. A reusable investigation core with child safety as its first specialization, connected through stable contracts and a registration seam; an explainable scoring engine decoupled from data collection so scores are reproducible and traceable; and a web layer that reads the same persisted data the command-line tools produce, so the dashboard and the reports always tell the same story.

Interface. A full command-line tool for investigators — including first-class commands for correlation, entity proposal and promotion, the identity graph, the timeline, account state and reconstitution chains, STIX export, and ATT&CK mapping — plus a session-gated web dashboard for review and visualization with risk-timeline, tier gauge, and explainable breakdown views.

What's Next

The platform build is complete: all four platform phases have shipped, along with the correlation CLI wiring, the persistence layer and its command-line surface, and a tokenless Discord public-OSINT module on the plugin contract. Remaining directions are additive:

Persistence in the dashboard. Reconstitution chains are available from the command line; surfacing them in the web interface, alongside automatic capture of removal and creation timestamps during collection, is the next increment.

Model-assisted behavioral analysis. Evaluating small, fine-tunable language models as an additional signal feeding the structured engine, kept as an explainable component rather than an opaque score.

Deeper platform coverage. Extending collection and correlation to additional platforms central to how minors socialize online, each built on the existing plugin contract with public data only.

PhotoDNA / NCMEC activation. The CSAM hash-detection architecture is complete and tested; activation waits on external authorization, exactly as the governance framework requires.

Collaboration. Engaging with child-safety organizations and platform Trust & Safety teams that build and use tooling in this space.

Summary

WhisperWard demonstrates the ability to take a hard, sensitive problem and build a complete, responsible, production-grade solution — and then to evolve it like a platform: robust data collection, transparent risk assessment, analyst-gated identity resolution, measurable enforcement outcomes, standards-aware intelligence export, forensic evidence handling, and strong ethical guardrails. It is public, deployed, thoroughly tested, and built to the standard this work demands.

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